

Software Requirements Specification

AI-Powered Mobile Fitness Application

Document Status: Version 1.0 (Draft)	Team Name: Trauma Squad (Team 23)
SDLC Methodology: Agile / Scrum	Primary Platform: Mobile App (iOS / Android)
Core Stack: Vision Pose AI, Local LLM, Personal Model	Primary Roles: User, Trainer, Administrator

1. Introduction

1.1 Purpose and Scope

Exercising independently at home often leads to incorrect posture, unmonitored muscle fatigue, and ineffective routine planning due to the lack of real-time personalized feedback. The **AI Fitness Application** is a mobile intelligent workout assistant designed to bridge this gap by utilizing the smartphone camera, computer vision pose estimation, and on-device machine learning to track body landmarks, evaluate movement form in near-real-time, and monitor muscle fatigue and recovery status. Powered by a Local LLM acting as an AI Coach, the application provides adaptive workout planning, daily goal management, and interactive guidance based on individual performance history stored in a central Personal Model. To ensure safety and human expertise, high-risk or low-confidence AI recommendations are escalated to a dedicated Trainer Platform for human review before execution. The Version 1.0 MVP focuses on core pose estimation, exercise form scoring, fatigue/recovery modeling, personal tracking, LLM-based coaching, and human trainer workflows. Success is measured by enabling users to execute workout routines safely with accurate real-time feedback while preventing overtraining and injury.

1.2 Users and Stakeholders

Name / Role	Kind	What They Need	What They Fear / Forbid
User	Primary User	Register account, build physical profile, receive real-time posture feedback during workouts, track muscle fatigue and recovery, receive adaptive workout plans, and chat with AI Coach.	Receiving inaccurate or overly confident posture feedback; performing exercises with unsafe form; privacy violations involving raw camera feeds.
Trainer	Secondary Stakeholder / Human Reviewer	Access assigned users' workout history, form analysis, and muscle status; review, approve, modify, or reject AI recommendations; issue custom instructions.	Accessing data of unassigned users; AI publishing high-risk workout plans without human review.
Administrator	System Administrator	Manage user accounts, role-based access control (RBAC), system configurations, logs, AI model versions, and exercise database entries.	Unauthorized configuration changes; unaudited system access; privacy breaches of sensitive user logs.

2. Overall Description

2.1 Product Context — In Scope vs Out of Scope

IN SCOPE (MVP Version 1.0)	OUT OF SCOPE (Explicit Promise)
• User account creation, authentication, and profile tracking • Device camera access and quality validation • Human body landmark pose estimation and tracking • Exercise recognition and repetition/set counting • Real-time exercise form analysis and corrective feedback • Muscle group workload estimation • Muscle fatigue scoring and recovery status tracking • Personal Model context tracking and adaptation • Local LLM AI Coach for workout planning and daily goals • Calorie expenditure and hydration estimation engines • Conversational AI Chatbot interface • Human-in-the-Loop Trainer Review Platform • AI Recommendation history, safety layer, and explainability	• Medical disease diagnosis or physical injury treatment • Replacing certified medical doctors or physical therapists • Formal medical certification of fitness outcomes • In-app payment processing or subscription features • Wearable hardware integration (e.g., smartwatch sensors) • In-app social media feeds or public marketplaces

2.2 Assumptions and Constraints

- **Assumption:** Users possess a smartphone with a functional camera running supported iOS or Android operating systems with internet connectivity for backend synchronization.
- **Constraint:** Camera pose estimation and form analysis must execute locally on-device or within strict latency budgets to maintain near-real-time feedback.
- **Constraint:** Local LLM execution relies on structured input/output JSON schemas and cannot directly mutate backend database states without validation.
- **Constraint (Privacy):** Raw video feeds are processed in memory for pose landmark extraction and must never be stored permanently or transmitted off-device without explicit consent.
- **Constraint:** Development schedule bounded by the project milestone submission window.

3. Functional Requirements

FR-1 — User Management

- User Story:** As a User, I want to create and manage my account profile, so that the application can tailor workout routines and tracking to my physical metrics and fitness level.
- Pain this traces to:** Fitness recommendations fail when user physical attributes, experience levels, and historical constraints are missing or scattered.
- Acceptance Criteria:**
1. Given valid user details (age, height, weight, activity level, goals), when submitted during registration or profile update, then the system saves the profile and updates the Personal Model context.
 2. Given an unauthenticated request to access user profile or workout features, when processed, then the system rejects access and prompts authentication.

FR-2 — Camera AI and Pose Estimation

User Story: As a User, I want the system to access my smartphone camera and track my body pose landmarks during a workout, so that my movement technique can be analyzed in real time.

Pain this traces to: Users struggle to position themselves correctly for automated workout tracking without clear feedback on camera visibility.

Acceptance Criteria:

1. Given camera permissions granted, when a workout starts, then the system captures video frames, detects key human body landmarks, and tracks them continuously.
2. Given poor lighting or body occlusion where pose confidence falls below the minimal threshold, then the system displays a warning and suppresses ungrounded feedback.

FR-3 — Exercise Recognition and Repetition Counting

User Story: As a User, I want the application to automatically identify supported exercises and count repetitions and sets, so that I can focus entirely on executing my workout.

Pain this traces to: Manual repetition logging during intense workouts distracts users and leads to incomplete tracking logs.

Acceptance Criteria:

1. Given a user performing a supported exercise (e.g., Squat), when repetition state transitions are completed, then the system accurately identifies the exercise and increments rep and set counts.
2. Given an unsupported movement or transition state, when detected, then the system maintains current set state without generating false rep counts.

FR-4 — Exercise Form Analysis

User Story: As a User, I want real-time posture analysis and corrective guidance, so that I can correct faulty technique and prevent physical injury.

Pain this traces to: Trainees perform exercises with improper form without realizing specific joint misalignments or movement errors.

Acceptance Criteria:

1. Given a completed rep, when posture errors are detected (e.g., knee caving in squats), then the system displays a Form Score, confidence rating, specific error tags, and actionable verbal guidance.
2. Given correct posture execution, when evaluated, then the system displays a high Form Score and logs the result to the workout session record.

FR-5 — Muscle Group Analysis

User Story: As a User, I want to view targeted primary and secondary muscle groups for each exercise, so that I understand my training workload distribution.

Pain this traces to: Users lack insight into which specific muscle groups absorb workload during complex multi-joint movements.

Acceptance Criteria:

1. Given a recorded workout session, when analyzed, then the system displays primary and secondary muscle groups with estimated activation levels (e.g., Quadriceps: High, Gluteus: High).
2. Given muscle activation results, when presented on the UI, then values are explicitly labeled as estimated workload rather than direct medical physiological measurements.

FR-6 — Muscle Fatigue and Recovery Estimation

User Story: As a User, I want estimated muscle fatigue and recovery indicators, so that I can decide which muscle groups require rest or lighter training.

Pain this traces to: Trainees repeatedly train fatigued muscle groups, causing overtraining syndrome and chronic muscle strain.

Acceptance Criteria:

1. Given historical workout volume, performance trends, and user feedback, when calculated, then the system categorizes muscle groups into recovery states (e.g., High Fatigue -> Rest Recommended) with underlying rationale provided.
2. Given fatigue and recovery statuses, when displayed, then recommendations are explicitly presented as wellness recovery estimates, not medical diagnoses.

FR-7 — Personal Model Context Tracking

User Story: As a User, I want my workout history, performance metrics, and recovery states retained in a Personal Model, so that future guidance adapts continuously to my progress.

Pain this traces to: Standard fitness apps treat workouts in isolation, ignoring cumulative fatigue and long-term user progression.

Acceptance Criteria:

1. Given completed session metrics (reps, form scores, fatigue scores), when a workout ends, then the system updates the user's central Personal Model with exact timestamps.
2. Given subsequent workout plan requests, when processed, then the system references historical Personal Model trends to adapt training recommendations.

FR-8 — AI Coach and Local LLM Integration

User Story: As a User, I want the Local LLM AI Coach to generate and adjust workout plans based on my Personal Model, so that my routines remain aligned with my goals and recovery status.

Pain this traces to: Static workout plans fail when users encounter unforeseen fatigue, missed days, or shifting personal goals.

Acceptance Criteria:

1. Given personal context (goals, history, recovery status), when requested, then the Local LLM generates structured daily goals and adaptive workout plans accompanied by natural explanations.
2. Given an LLM workout recommendation, when generated, then it must pass through the system Safety Layer before reaching the user and cannot directly mutate database records.

FR-9 — Calorie and Hydration Estimation

User Story: As a User, I want estimated calorie expenditure and hydration recommendations, so that I can manage energy output and fluid recovery.

Pain this traces to: Users find it difficult to estimate energy output and fluid replacement required after strenuous training sessions.

Acceptance Criteria:

1. Given user biometric profile and workout parameters (duration, intensity, reps), when computed by the Calculation Engine, then estimated calories and fluid requirements are rendered.
2. Given Local LLM text generation, when presenting nutrition or calorie figures, then the LLM must consume values calculated by the Calculation Engine rather than generating invented numbers.

FR-10 — AI Chatbot Interface

User Story: As a User, I want an interactive conversational AI Chatbot, so that I can ask specific questions about my workout plan and recovery options.

Pain this traces to: Trainees have specific questions about exercise choices but lack immediate access to a personal assistant.

Acceptance Criteria:

1. Given user inquiries regarding plans or recovery, when submitted, then the Chatbot provides contextual responses using the authorized Personal Model data while flagging out-of-scope queries.
2. Given high-risk or ambiguous user messages (e.g., pain reports), when detected, then the Chatbot escalates the interaction to the Trainer Review queue.

FR-11 — Trainer Platform (Human-in-the-Loop)

User Story: As a Trainer, I want to review, modify, approve, or reject AI-generated recommendations for users under my care, so that human expertise oversees automated recommendations.

Pain this traces to: Fully automated AI workout plans can produce unsuitable recommendations for individuals requiring professional human oversight.

Acceptance Criteria:

1. Given an AI recommendation flagged for human review, when accessed by the assigned trainer, then the trainer can inspect user context and approve, modify, or reject the plan with optional commentary.
2. Given a trainer review action, when saved, then the decision is recorded with a timestamp and trainer ID in the audit log, and the updated plan is delivered to the user.

FR-12 — Safety Requirements and Exception Handling

User Story: As a User, I want the system to enforce safety thresholds and escalate high-risk scenarios, so that AI outputs never compromise physical safety.

Pain this traces to: Autonomous AI models might suggest excessive physical loads when user pain or severe fatigue is present.

Acceptance Criteria:

1. Given user-reported pain symptoms or severe fatigue indicators, when triggered, then the system halts automated workout escalation and routes the user context to Human Trainer Review.
2. Given insufficient camera confidence or missing data context, when processing, then the system restricts autonomous exercise recommendations and notifies the user.

4. Non-Functional Requirements

ID	Kind	Statement (Testable)	How We Will Check
NFR-1	Performance	Real-time posture feedback latency shall not exceed 100 ms between frame capture and UI rendering.	Timestamp logging from camera frame capture to UI render across 50 test workout trials.
NFR-2	Performance	On-device pose estimation inference latency shall remain under 50 ms per frame on reference devices.	Model execution benchmark timing across 1,000 continuous video frames.
NFR-3	Performance	Backend API requests shall return responses within 1.5 seconds under standard network conditions.	Automated API test suite (e.g., Postman/ K6) measuring latency across 100 concurrent requests.

ID	Kind	Statement (Testable)	How We Will Check
NFR-4	Reliability	The system shall implement graceful degradation to basic rep counting if form model inference or network fails.	Network disconnect and sub-model fault injection during active workout sessions.
NFR-5	Usability	A first-time user shall be able to start a camera-guided workout session within 45 seconds of app launch.	Timed usability trial with 5 novice participants on fresh installations.
NFR-6	Usability	Real-time exercise form corrections shall use clear action phrasing under 15 words.	Content audit of feedback template database against usability standards.
NFR-7	Security	The system shall authenticate all requests using secure OAuth2/JWT tokens and reject unauthorized calls.	Penetration test attempting unauthenticated access to backend endpoints.
NFR-8	Security REQ.	A user or trainer shall access only authorized profile/workout records; requesting another user's ID directly shall fail.	Modify userId parameter in API requests and verify HTTP 403 Forbidden rejection response.
NFR-9	Security	All data transmitted between client mobile app and server endpoints shall be encrypted using TLS 1.3.	Wireshark packet capture audit during synchronization to verify transport encryption.
NFR-10	Privacy	Camera processing shall extract pose landmarks locally on-device; raw video frames shall not be transmitted without explicit consent.	Network payload audit verifying no raw video or image payloads leave the device.
NFR-11	Privacy	The system shall enforce data retention policies that automatically purge raw session telemetry older than 90 days.	Automated database retention script execution audit in staging environment.

5. Use Cases

5.1 Use-Case List

Use Case	Actor	Related FR	Brief Success Story
UC-1: Start Camera Workout	User	FR-2, FR-3, FR-4, FR-5	User selects workout → Camera initializes → System verifies lighting/pose → Real-time form feedback rendered → Reps counted → Personal Model updated.
UC-2: Consult AI Coach	User	FR-8, FR-10, FR-12	User asks training question → Chatbot fetches Personal Model & recovery status → Local LLM generates response → Safety Layer validates → Answer delivered.
UC-3: Review AI Recommendation	Trainer	FR-11, FR-12	AI flags plan due to high fatigue → Trainer reviews user history → Trainer modifies plan → Audit recorded → Approved plan sent to user.

5.2 Use-Case Diagram Note

Note: UML Actor/Association and <<include>> diagrams are deferred to Milestone 3 / Milestone 4 as per course curriculum guidelines. Use case mapping above establishes the functional scope.

6. Out of Scope

1. Medical disease diagnosis, physical injury treatment, or replacing certified physical therapists.
2. Formal medical certification or legal guarantee of physical outcome.
3. In-app payment processing, paid subscriptions, or e-commerce features.
4. Hardware wearable integration (e.g., smartwatch heart rate sensors) — deferred to Future Scope.
5. Social media feeds, community interaction, or public trainer marketplaces.

7. Traceability Matrix (Golden Thread)

Problem / Pain	Requirement (FR)	Solution / Design	Feature Built	Test
Users perform workouts with poor posture without realizing error points	FR-4 Form Analysis	Computer Vision Pose Estimation + Form Classifier	Real-time posture feedback & Form Score	Form error test scenarios using synthetic and live pose datasets
Trainees suffer overtraining due to lack of muscle fatigue visibility	FR-6 Fatigue & Recovery	Workload & Muscle Recovery Calculation Engine	Muscle recovery dashboard & rest recommendations	Integration tests evaluating fatigue buildup across multi-session logs
Generic static plans fail to adapt when users encounter fatigue or missed days	FR-8 AI Coach (LLM)	Local LLM adaptive planning with Personal Model context	Adaptive daily goal and workout generator	Integration test validating LLM prompt context framing and Safety Layer filters
Pure AI automation may issue unsafe workouts to high-risk trainees	FR-11 Trainer Review	Human-in-the-Loop approval queue	Trainer review workflow portal	Role-based authorization and state transition unit tests

8. AI Usage Log

Date	Tool / Model	What We Asked	What AI Produced	What We Changed / Verified
28 Aug 2026	ChatGPT / GPT-5.6	Reformat AI Fitness SRS to match StudyMate SRS structure without altering underlying requirement scope.	Structured draft with User Stories, Given/When/Then ACs, NFR test methods, and Golden Thread matrix.	Removed draft annotations (ORIGINAL CONTENT, ADDED); translated all Thai prose to professional English; consolidated duplicate SHALL lists into acceptance criteria.

9. Self-Check

- ✓ **Scope and out-of-scope are explicitly stated:** Defined in Section 2.1 and Section 6.
- ✓ **FRs are organized as testable User Stories:** Every FR includes As a... I want... so that... linked to a real user pain point.
- ✓ **Acceptance Criteria are testable:** Every FR features numbered Given... when... then... scenarios.
- ✓ **NFRs are measurable and include check methods:** Section 4 contains quantified targets and explicit How we will check procedures.

- ✓ **Use-case list matches main workflows:** Section 5 maps UC-1 through UC-3 directly to functional requirements.
- ✓ **Golden Thread matrix is present:** Section 7 traces Pain → Requirement → Solution → Feature → Test.
- ✓ **AI usage log is honest and accurate:** Section 8 documents prompt history, outputs, human edits, and key takeaways.